Fundamentals of excel for data driven decision making

About this training

- This training is aimed at equipping the participants from beginner friendly introductions to more advanced excel techniques for their day to day data analytics for data driven decision making. It contains step-by-step comprehensive guide on performing the analytics. It also contains easy to follow hands on exercises.

Learning objectives

By the end of this training, the participants should be able to:

1. Identify the MS Excel tools for data analysis
2. Use MS Excel tools for data analysis
3. Use MS Excel tools for data visualization

Training Outline

Introduction to Data analysis

- Process of Data analysis
- Methods of Data analysis

Importance of data analysis

Excel Basics

- Data Entry, Editing and Number Formatting

Excel formulas and functions for data analysis

- Excel formulas and functions

Power Query

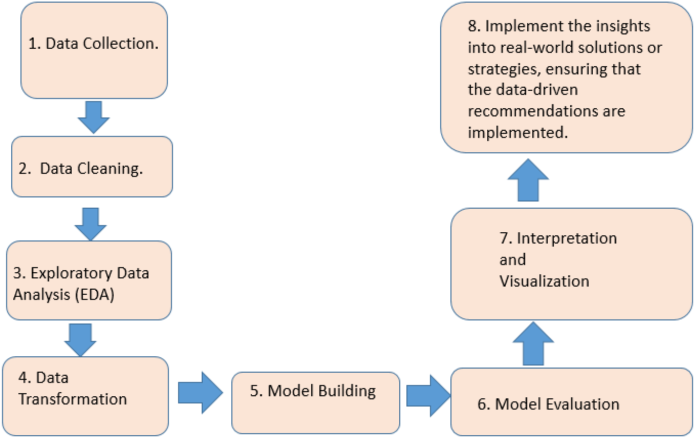
Pivot Tables

Data Visualization in Excel

- Excel Charting and Pivot Charts
- Dash Boards

Introduction to Data Analysis

The data analysis process is a structured sequence of steps that lead from raw data to actionable insights.



Importance of data analysis

- **Uncovering Patterns and Trends:** Data analysis allows researchers to identify patterns, trends, and relationships within the data.
- **Testing Hypotheses:** Research often involves formulating hypotheses and testing them. Data analysis provides the means to evaluate hypotheses rigorously.
- **Making Informed Conclusions:** Data analysis helps researchers draw meaningful and evidence-based conclusions from their research findings. It provides a quantitative basis for making claims and recommendations.
- **Enhancing Data Quality:** Data analysis includes data cleaning and validation processes that improve the quality and reliability of the dataset.
- **Supporting Decision-Making:** In applied research, data analysis assists decision-makers in various sectors, such as business, government, and healthcare. Policy decisions, marketing strategies, and resource allocations are often based on research findings.
- **Identifying Outliers and Anomalies:** Outliers and anomalies in data can hold valuable information or indicate errors. Data analysis techniques can help identify these exceptional cases

Let's begin with Excel

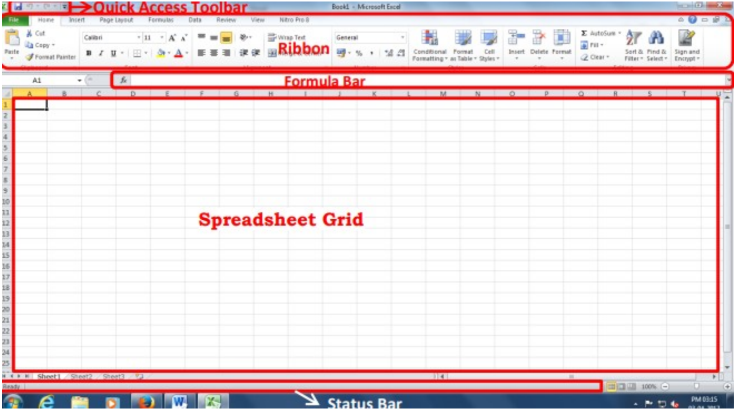
- **Introduction to Ms Excel**
- Microsoft Excel is the most widely used spreadsheet, and is available within the Microsoft Office suite of programs.
- MS Excel is used for making spreadsheets.
- The power of a spreadsheet comes from the ability to enter formula into the cells of the spreadsheet so that results can be calculated automatically.
- This allows the contents of any cell that contains a formula to be determined automatically from the contents of other cells.
- Furthermore, the calculation is dynamic in the sense that the result of the formula changes automatically whenever the values that the formula changes.

To get into Excel;

1) Double click on the Microsoft Excel icon if there is one on the computer desktop.

2) Alternatively, click on the Start button in the bottom left corner of the screen, move the cursor to Programs to open up the programs menu, and then click on Microsoft Excel.

This will launch excel and below is the excel window you shall see:



Excel Spreadsheet, Workbook or File Object

- An excel file also called a workbook contains many components.
- There are several data sheets called work sheets in a workbook.
- The user can add or delete these worksheets.
- Each worksheet has a unique name and a workbook MUST have at least one sheet.
- Each worksheet contains rows and columns that combine to provide the gridlike structure of the worksheet.
- The rows in the worksheet are numbered from 1,2,3,... while the columns are represented in alphabets A,B,C...

- **Cells**

Within the worksheet, there are numerous cells referenced by the column then row e.g. B23 is the second column and 23rd row.

- **Range**
- Range is a group of cells or a group of rows and columns.
- A range is written as first cell:last cell e.g. A2:AB22 is from cell A2 to cell AB22.
- A range could also be the union of multiple blocks of cell written as (A2:B6,A7:B15)
- **Cell references**
- All formulas, functions, charts and other features reference the data in excel by referencing the cell and not the data itself. e.g A3

- **Renaming sheets**
- Excel provides default names to sheets in a workbook and the sheets are named as “sheet1”, “sheet2”,. . . “sheetN”.
- As the user, you might want to rename the sheets to some name that represents the content of the worksheet.

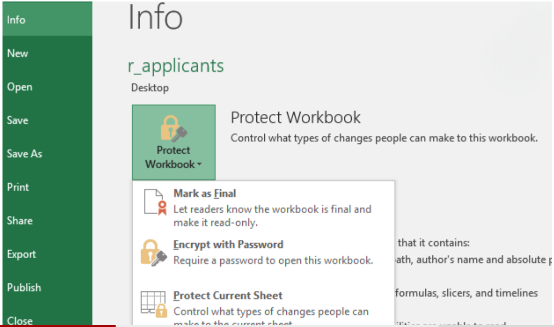
To rename the sheet;

1) Double click on the sheet name and the sheet name tab will be highlighted.
2) In this highlighted area, delete the old sheet name and write the new sheet name.

- **Protecting sheets against unauthorized changes/viewing by others**
- You can either protect the entire workbook or just specific worksheets from being viewed or changed by other people.
- To achieve this;

1) go to files menu -> Info -> protect workbook ->encrypt with password.
2) Enter the password that will be required to access the workbook.

- To protect a particular sheet, right click on the sheet you want to protect, select **protect sheet** option, select password option.



Data Entry, Data Editing and Formatting

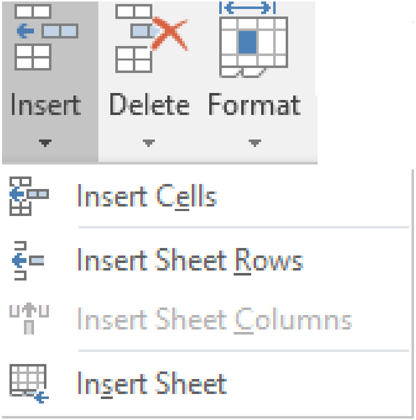
To enter data in the formula bar, click in the cell that you want the formula in and type in the data.

- **Modifying a worksheet**
- Excel allows you to move, copy, and paste cell content through cutting and pasting and copying and pasting.
- This can be done through right-clicking on the cell of interest or selecting the range of cells or row or column.

NB: When you copy or cut the contents of a cell, excel copies or moves the entire cell including the formulas, resulting values, cell formats and the comments if any.

Adding or deleting a row and column

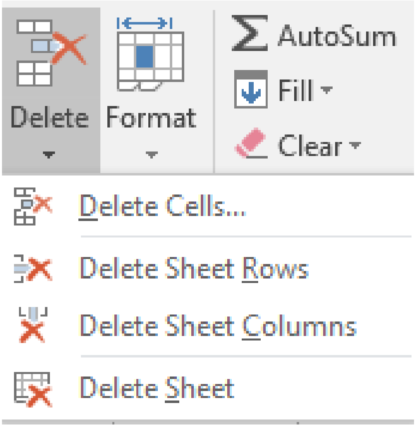
To insert a new row or column;
1) Right click on the row or cell of interest
1) For the column, Right click on the column letter of interest
2) Click Insert
2) Select the option of Insert
alternatively;
1) Place the cursor below where you want to add the new row
2) Click the Insert button on the Cells group of the Home tab
3) Click the appropriate choice: Cell, Row, or Column



Note that excel always adds a new row above the row that was selected.

Note that excel always adds a new column to the LEFT of the column that was selected.

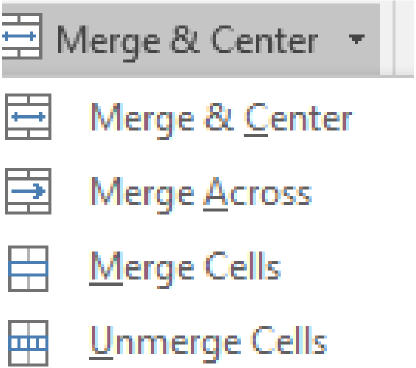
To delete a row;
1) Right click on the row of interest or column or cell
2) Click Delete



Merge or split cells

Merging of adjacent cells combines several cells to become one large cell

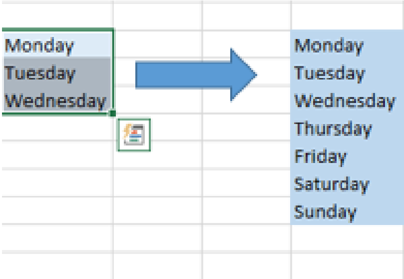
To merge and center cells;
1) Select two or more adjacent cells that you want to merge.
2) On the Home tab, in the Alignment group, click Merge and Center.
3) The cells will be merged in a row or column, and the cell contents will be centered in the merged cell.
To split the merged cells;
1) Select the merged cell you want to split.
2) To split the merged cell, click Merge and Center.
3) Click on the Unmerge cells, the cells will split.



Auto Fill

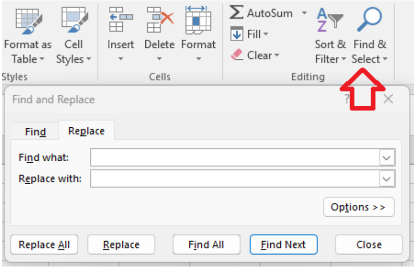
- The Auto Fill feature automatically fills cell data or series of data in a worksheet into a selected range of cells. For example, Days of the week or months of the year.

To use the Auto fill feature;
1) Fill in the first 2 cells of the series
2) Click on the Fill handle
3) Drag the Fill handle to automatically fill the cells.



Find or Replace data

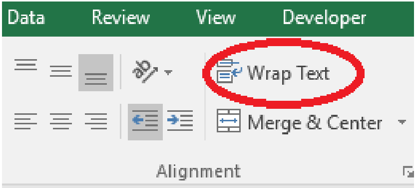
To find data or find and replace data;
1) Click the Find & Select button on the Editing group of the Home tab
2) Choose Find or Replace
3) Complete the Find What text box
4) Click on Options for more search options



Wrap Text

This enables you to display multiple lines of text inside one cell.

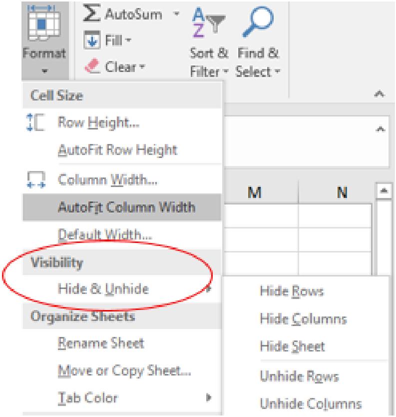
To wrap text;
1) Click the cell in which you want to wrap the text.
2) On the Home tab, in the Alignment group, click Wrap Text.
3) The text in your cell will be wrapped.



Hide or Display rows and columns

You can hide a row or column by using the Hide command or when you change its row height or column width to 0 (zero). You can display either again by using the Unhide command.

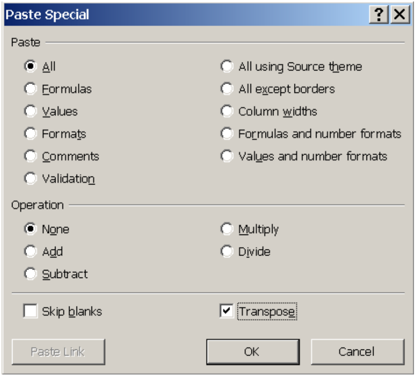
To Hide Rows or Columns;
1) Select the rows or columns that you want to hide.
2) On the Home tab, in the Cells group, click Format.
3) Under Visibility, point to Hide & Unhide, and then click Hide Rows or Hide Columns.
To Unhide Rows or Columns;
1) Select the rows, columns or entire sheet to unhide.
2) On the Home tab, in the Cells group, click Format.
3) Under Visibility, click Unhide Rows or Unhide Columns.



Transpose

Sometimes data is mistakenly entered with an unsuitable layout. Tables can be turned around to make row headings into column headings, thus allowing them to be more readable and functional as databases.

To turn the data around is to:
1) Select the table
2) Copy
3) click on another area of the sheet (or another sheet)
4) Paste Special
5) check the Transpose box
6) OK



Excel formulas and functions for data analysis

- **Excel Formulas**
- Formulas are what make a spreadsheet preferrable, useful and easy to use.
- A formula is a set of mathematical instructions that can be used in Excel to perform calculations.
- Copying of the formulas makes excel quick and easy to manage.
- A formula begins with $=$ sign.

There are many components of a formula in excel:

- The reference; which is the cell or range of cells that you want to use in your calculations,
- The operator that specifies the exact operation you want performed,
- The constants which are the numbers or text values that do not change.

The common arithmetic operators are:

Table 1: Common Arithmetic Operators for Formulas

Operator	Example
Addition(+)	A3+C5
Subtraction(-)	A5-A7
Multiplication(*)	C3*D2
Division(/)	F2/C2
Exponentiation(^)	F3^2

- **Relative Reference and Absolute Reference**
- By default, formulas in Excel use relative cell references. This means that, as a formula is copied and pasted to other cells, the cell references in the formula change to reflect the formula's new location.
- On the other hand, absolute cell reference in a formula does not change when the formula is copied and pasted into other cells.

For example, to make the reference to cell A3 absolute, and therefore ensure that it remains unchanged when it is copied to other cells, you include $ signs (A3).

- **Excel Functions**

These are pre-defined or built in formulas in excel. They also begin with = sign. To insert a function;

1) click the cell where you want the function applied,

2) Click the insert button (fx)

3) Then choose the appropriate function. Ensure that the desired range is selected.

alternatively,

- Using the Autosum function on the ribbon, select the appropriate function.

Common Excel Statistical Functions

- **SUM Function**: This gives the total or aggregate values.

Example:

- SUM(C2:H2) – A simple selection that sums the values of a row.
- SUM(A2:A8) – A simple selection that sums the values of a column.
- SUM(A2:A7, A9, A12:A15) – A sophisticated collection that sums values from range A2 to A7, skips A8, adds A9, jumps A10 and A11, then finally adds from A12 to A15.
- SUM(A2:A8)/20 – Shows you can also turn your function into a formula.

- **MIN Function**: This function gives the smallest value in a given range

Example:

- MIN(B2:C11)
- MIN(C1,D1,A3,B2) — This Excel function determines the smallest value among the 4 cells listed

- **MAX Function**: This function gives the largest value in a given range

Example:

- MAX(B2:C11)
- MAX(A1:A9,300) — This Excel function determines the largest value among the numbers in cells A1 through A9, and the number 300.

- **AVERAGE Function**: This function gives the average value in a given range.
- The syntax is given by:

AVERAGE(number1, [number2], . . .)

Example:

AVERAGE(A1:A10)

- **COUNT Function**: This function counts the number of cells containing numerical values in a given range. The syntax is given by:

COUNT(value1, [value2], . . .)

Example:

- COUNT(B:B) – Counts all values that are numerical in B column.
- COUNT(C1:E1) – Now it can count rows.
- COUNT(A3:A7,10) - The number of cells with numeric values in cells in A3 to A7, plus 1 for the number 10.

- **COUNTIF Function**: This function allows you to display the number of cells in a single range whose values meets specific criteria. The syntax is given by:

COUNTIF(range,criteria)

Example:

- COUNTIF(C1:C10,3) counts the number of cells in C1-C10 whose value equals 3
- COUNTIF(C1:C10,"no") counts the number of cells in C1-C10 that contain the word no
- **COUNTA Function**:The number of non-empty cells.
- It counts all cells regardless of type.
- That is, unlike COUNT that relies on only numerics, it also counts dates, times, strings, logical values, errors, empty string, or text.
- **COUNTBLANK Function**:The number of blank cells.
- It counts all the blank cells within a given range.

- **IF Function**: It gives formulas decision making capacities.
- The IF function includes a logical test, that compares a cell against a specified value and produces an answer of either TRUE or FALSE. The syntax is:

IF(Logical Test, Value 1, Value 2) which produces these results:

If the Logical Test is TRUE, Value 1 is inserted into the cell, If the Logical Test is FALSE, Value 2 is inserted into the cell

Example:

- IF(D1>16,13,14) - This Excel function checks to see if the value in D1 is greater than 16. If so, Excel returns a value of 13. If not, a value of 14 is returned.
- IF(D1<500,"Poor performance"," within Budget") -If value in D1 is less than 500,displays Poor performance. If not, displays within Budget.
- **Nested IF**

This is When IF functions are used inside of IF functions.

Example:

IF(B9>=18,"Adult",IF(B9>12,"Teen","Child"))

Text Functions

- **Capitalization** using UPPER functions-All letters will in Capital, LOWER functions-All letters will be in lower case.

Example:

- UPPER(B2)- All letters are capitalized.
- LOWER(B3)- All letters are not capitalized.
- PROPER(A3)- Every first letter of each Word should be Capital.
- TRIM function- for removing unwanted spaces in words e.g. TRIM(A2)
- **LEN function-** for calculating total number of characters in a word including spaces
- **CONCATENATE function-** This is a function that is used for joining two or more strings or columns.
- It comes in handy when dealing with a large dataset.

Example:

- CONCATENATE(B3,“;”;D4)

Reference Functions

LookUp functions

- The lookup function returns a value from a table by looking up another related value.
- The **Vlookup function** searches for a value in a first column of a table and then returning the corresponding value in another table.
- It is a vertical lookup meaning that it searches for values in columns. The syntax is:
- VLOOKUP(lookup value, range containing the lookup value, the column number in the range containing the return value, Approximate match (TRUE) or Exact match (FALSE)).

Example:

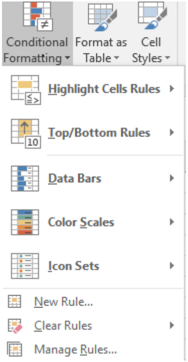
VLOOKUP(B11,D2:F7,2)

- The Hlookup searches for a value when the data is organized differently in row form.

Conditional Formatting

- Conditional Formatting allows you to change the appearance of a cell to highlight the important information in a cell.
- Changes can automatically be applied to: font colours/sizes, cell border/shading, etc. depending on certain conditions.

1) First you select (click and drag) the relevant column of data.
2) Then from the Home tab choose the Conditional Formatting button. Then choose Highlight Cells Rules, then choose an operation e.g. Less Than
3) Next, input a number above which the formatting will take effect, and select a style from the pull-down menu.
4) Choose OK



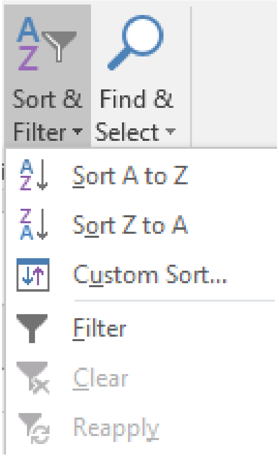
Sorting and Filtering

Sorting allows you to rearrange your data in either an ascending order or descending order. To execute a sort based on one column:

1) Highlight the cells that will be sorted
2) Click the Sort & Filter button on the Home tab
3) Click the Sort Ascending (A-Z) button or Sort Descending (Z-A) button

To sort on the basis of more than one column:

1) Click the Sort & Filter button on the Home tab
2) Choose which column you want to sort by first
3) Click Add Level
4) Choose the next column you want to sort
5) Click OK



Filtering allows you to display data that only meets a particular criteria.

To filter:

1) Click the column or columns that contain the data you wish to filter

2) On the Home tab, click on Sort & Filter

3) Click Filter button

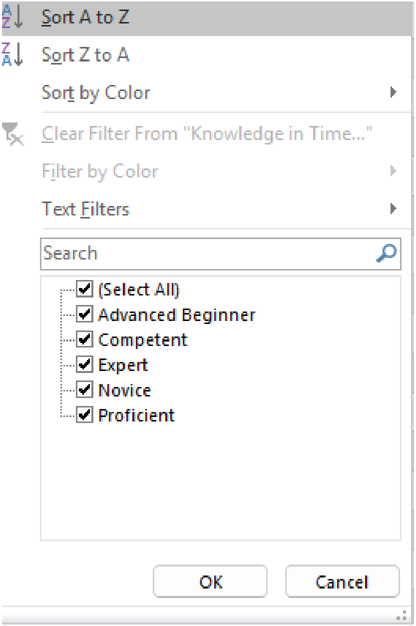
4) Click the Arrow at the bottom of the first cell

5) Click the Text Filter

6) Click the Words you wish to Filter

7) To clear the filter click the Sort & Filter button

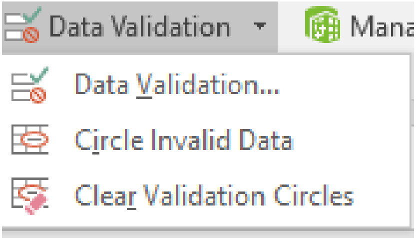
8) Click Clear



Data validation

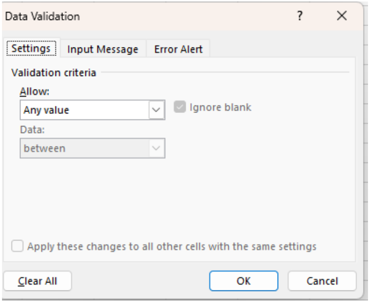
- Data validation is an important technique of ensuring that the data entered into excel only meets a particular criteria by restricting the data types and parameters on the data.
- It is possible in Excel to ensure that data can only be entered if it is of the type and within constraints set by you. Rogue values are therefore minimized.

To perform data validation:
1) When you have input the column heading
2) You can then highlight the column
3) Then choose Data Validation from the Data tab
4) Assign settings for the data that you will allow.
5) Note the checkbox; you have the option to ignore blank fields, or by unchecking the box, you would be accepting that the field may be empty of data.

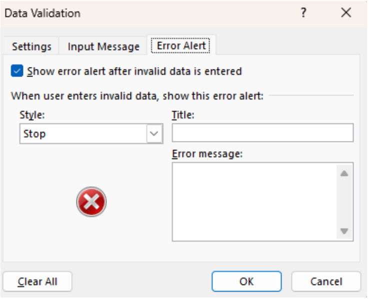


Note:

The Allow menu lets you select the data type allowed.
The Data menu lets you set parameters.



- An Input Message is a message which appears from a cell when it is selected.
- It may impart useful information to the person entering the data.
- To add an error message, click the Error Alert tab at the top of the Data Validation dialogue box



Pivot Tables

A Pivot Table is a way to present information in a report format.

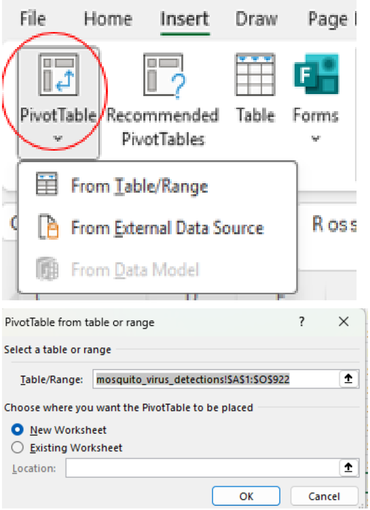
Creating a Pivot table

1) When you create a dynamic pivot table, each column of source data becomes a field that can be used in the pivot table.

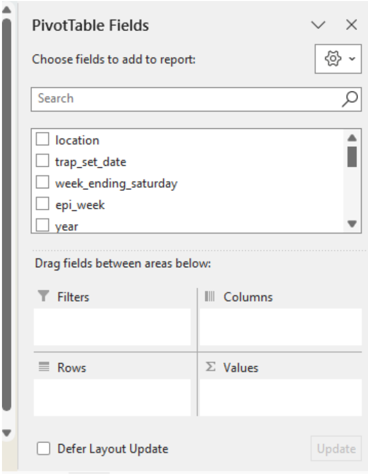
The fields summarise rows of information of the data source.

2) The names of the fields for the pivot table come from the titles of the columns of their data source.
3) Ensure that you have names for each column of the first row of the spreadsheet of the data source.
4) The data table must not have empty columns in the data that is used for the dynamic table.
5) Click and drag to select the dataset, including the titles of the columns.

Next, from the Insert tab, choose PivotTable.
6) Select a New Worksheet as the place where the Pivot Table will be placed. Click OK. Alternatively, a Pivot Table can be placed in an existing worksheet.



7) The table comes to life when you check field boxes on the PivotTable Field List.
8) Click on each field that you want to add to the Pivot Table, or drag and drop the fields to the header areas.
9) To see only data for Table A for example, you could set up the filters. Find them in the Row Labels pull-down menu.
9) From the filter menu, Select All to see all of the data again



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